

Krushna Lokare

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RESEARCH INTERESTS

- Analysis of nonlinear PDEs
- Phase-field models
- Numerical analysis and computational methods for PDEs, including neural-network-based schemes

EDUCATION

Indian Institute of Science Education and Research, Thiruvananthapuram (IISER TVM) *Nov 2022 – Present*
Integrated BS-MS in Mathematical Sciences

RESEARCH EXPERIENCE

Major Thesis: Nonlocal Cahn–Hilliard Models (ongoing) *May 2026 – Present*

Supervisor: Dr. Sheetal Dharmatti, School of Mathematics, IISER TVM

- Theoretical project on well-posedness, regularity, and long-time behaviour of Cahn–Hilliard, nonlocal Cahn–Hilliard, and Cahn–Hilliard–Navier–Stokes equations
- Studying nonlocal Cahn–Hilliard equations with singular potentials, the strict separation property, and convergence to equilibrium for logarithmic free energies
- Applying functional-analytic tools underlying well-posedness theory: energy methods, Sobolev estimates, compactness/fixed-point arguments

Physics-Informed Neural Networks for PDEs *Dec 2025 – Feb 2026*

Supervisor: Dr. Sheetal Dharmatti, School of Mathematics, IISER TVM

- Investigated residual-based PINN methods for nonlinear and higher-order PDE models, focusing on training behaviour and formulation choices

ENO and Neural-ENO Methods for Discontinuous Functions *May – Aug 2025*

Supervisor: Dr. Sheetal Dharmatti, School of Mathematics, IISER TVM

- Implemented ENO interpolation via Newton divided differences for piecewise-smooth functions; built adaptive stencil selection from third-order divided differences
- Recast ENO and ENO-SR schemes as ReLU neural networks, reproducing theoretical results
- Developed DeLENO, a classifier predicting stencil shifts, and compared ENO vs. DeLENO using total-variation error metrics

Undergraduate Research Intern, IISER TVM *Jan 2024 – May 2025*

Supervisor: Dr. Shyamal Ghosh, School of Data Science, IISER TVM

- Studied theoretical foundations of estimation, hypothesis testing, and reliability models using R
- Ran simulation studies for inference procedures in renewal and age-replacement models
- Reading project: goodness-of-fit test for MTTF in age-replacement models, including asymptotic behaviour and efficiency (PARE)

TECHNICAL SKILLS

- Programming: Python (NumPy, SciPy, Matplotlib, Pandas), R, MATLAB, C (HPC)

REFERENCE

Dr. Dharmatti Sheetal

Associate Professor (Mathematics)

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